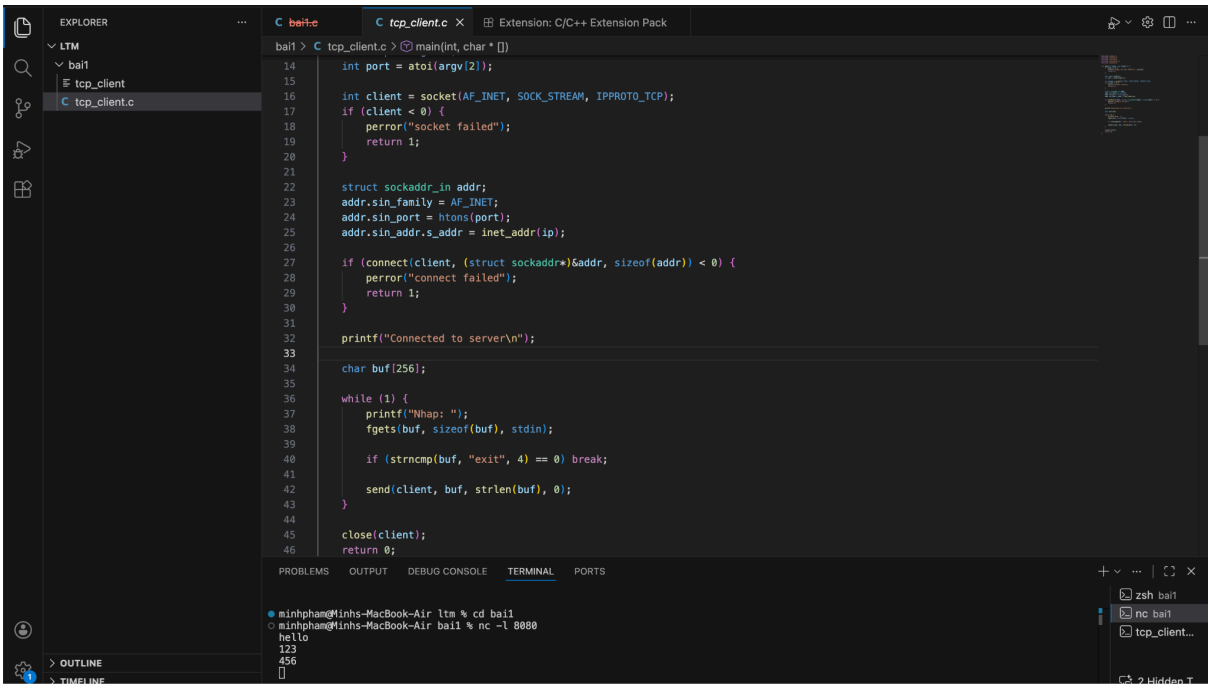
Báo cáo bài tập về nhà 17/3/2026

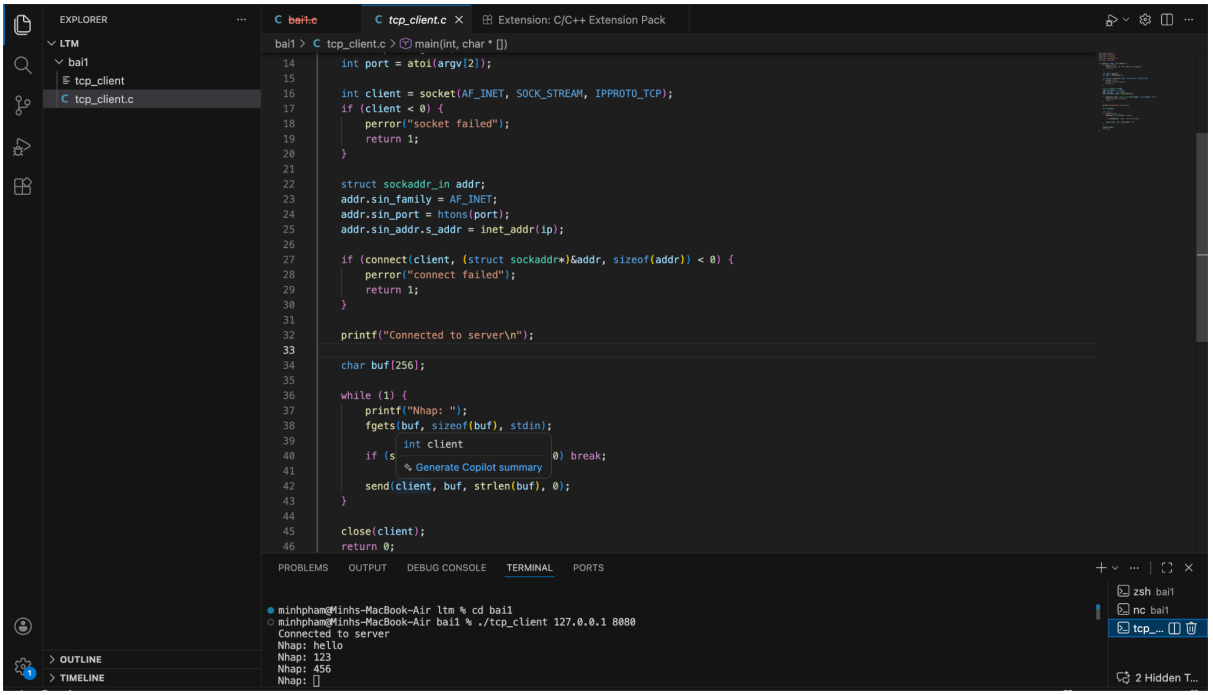
Họ và tên: Phạm Hoàng Minh

MSSV : 20235381

Mã lớp : 168503

Bài 1:





Bài 2:

EXPLORER

LTM
- bai1
  - tcp_client
  - tcp_client.c
- bai2
  - output.txt
  - tcp_server
  - tcp_server.c
  - welcome.txt

tcp_server.c | welcome.txt | output.txt | tcp_server | Extension: C/C++ Extension Pack

bai2 > tcp_server.c > main(int, char * [])

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <arpa/inet.h>
#include <unistd.h>

int main(int argc, char *argv[]) {
    if (argc != 4) {
        printf("Usage: %s <PORT> <WELCOME_FILE> <OUTPUT_FILE>\n", argv[0]);
        return 1;
    }

    int port = atoi(argv[1]);
    char *welcome_file = argv[2];
    char *output_file = argv[3];

    int listener = socket(AF_INET, SOCK_STREAM, IPPROTO_TCP);

    struct sockaddr_in addr;
    addr.sin_family = AF_INET;
    addr.sin_port = htons(port);
    addr.sin_addr.s_addr = INADDR_ANY;

    bind(listener, (struct sockaddr*)&addr, sizeof(addr));
    listen(listener, 5);

    printf("Server dang cho ket noi...\n");

```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
minhpham@Minhs-MacBook-Air ltm % cd bai2
minhpham@Minhs-MacBook-Air bai2 % gcc tcp_server.c -o tcp_server
minhpham@Minhs-MacBook-Air bai2 % ./tcp_server 8080 welcome.txt output.txt
Server dang cho ket noi...
Client connected
Nhan: hello
Nhan: abc
Nhan: 123
```

tcp_server... | tcp_client... | zsh bai2

OUTLINE

EXPLORER

LTM
- bai1
  - tcp_client
  - tcp_client.c
- bai2
  - output.txt
  - tcp_server
  - tcp_server.c
  - welcome.txt

tcp_server.c | welcome.txt | output.txt | tcp_server | Extension: C/C++ Extension Pack

bai2 > tcp_server.c > main(int, char * [])

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <arpa/inet.h>
#include <unistd.h>

int main(int argc, char *argv[]) {
    if (argc != 4) {
        printf("Usage: %s <PORT> <WELCOME_FILE> <OUTPUT_FILE>\n", argv[0]);
        return 1;
    }

    int port = atoi(argv[1]);
    char *welcome_file = argv[2];
    char *output_file = argv[3];

    int listener = socket(AF_INET, SOCK_STREAM, IPPROTO_TCP);

    struct sockaddr_in addr;
    addr.sin_family = AF_INET;
    addr.sin_port = htons(port);
    addr.sin_addr.s_addr = INADDR_ANY;

    bind(listener, (struct sockaddr*)&addr, sizeof(addr));
    listen(listener, 5);

    printf("Server dang cho ket noi...\n");

```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
minhpham@Minhs-MacBook-Air ltm % cd bai1
minhpham@Minhs-MacBook-Air bai1 % ./tcp_client 127.0.0.1 8080
Connected to server
Nhap: hello
Nhap: abc
Nhap: 123
Nhap:
```

tcp_server... | tcp_... | zsh bai2

OUTLINE

EXPLORER

LTM
- bai1
  - tcp_client
  - tcp_client.c
- bai2
  - output.txt
  - tcp_server
  - tcp_server.c
  - welcome.txt

tcp_server.c | welcome.txt | output.txt | tcp_server | Extension: C/C++ Extension Pack

bai2 > tcp_server.c > main(int, char * [])

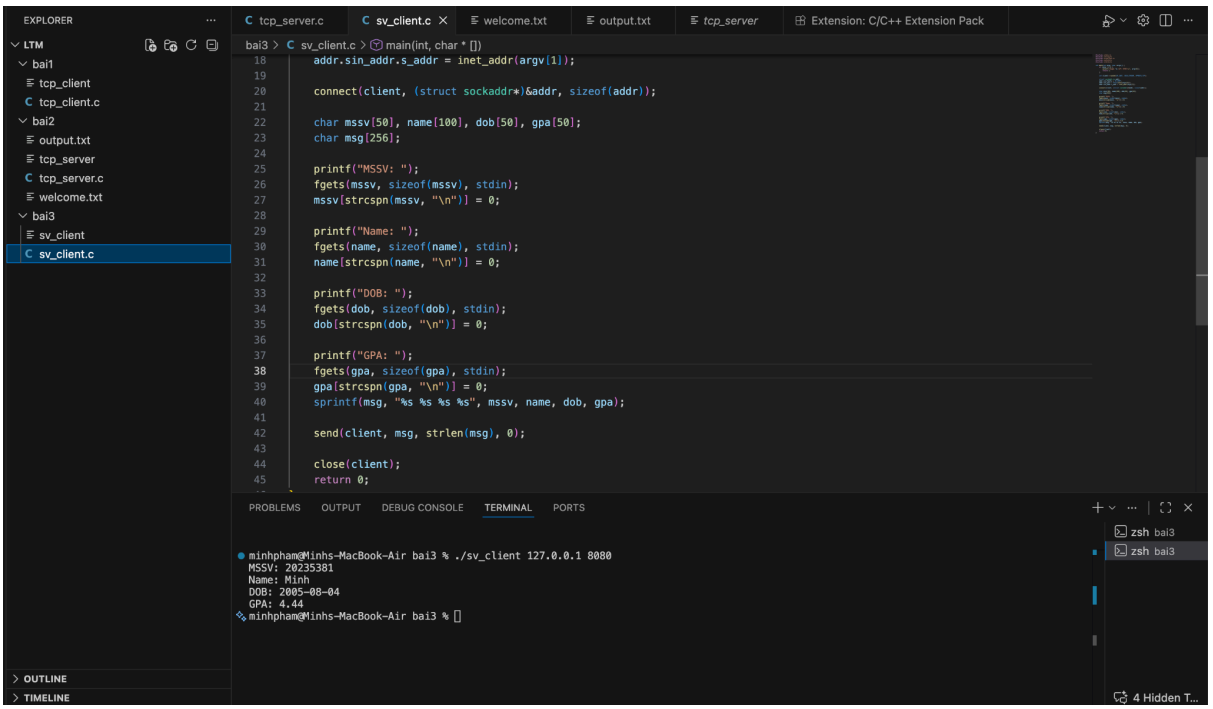
```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <arpa/inet.h>
#include <unistd.h>

int main(int argc, char *argv[]) {
    if (argc != 4) {
        printf("Usage: %s <PORT> <WELCOME_FILE> <OUTPUT_FILE>\n", argv[0]);
        return 1;
    }

    int port = atoi(argv[1]);
    char *welcome_file = argv[2];
    char *output_file = argv[3];

    int listener = socket(AF_INET, SOCK_STREAM, IPPROTO_TCP);

    struct sockaddr_in addr;
    addr.sin_family = AF_INET;
    addr.sin_port = htons(port);
    addr.sin_addr.s_addr = INADDR_ANY;

    bind(listener, (struct sockaddr*)&addr, sizeof(addr));
    listen(listener, 5);

    printf("Server dang cho ket noi...\n");

```

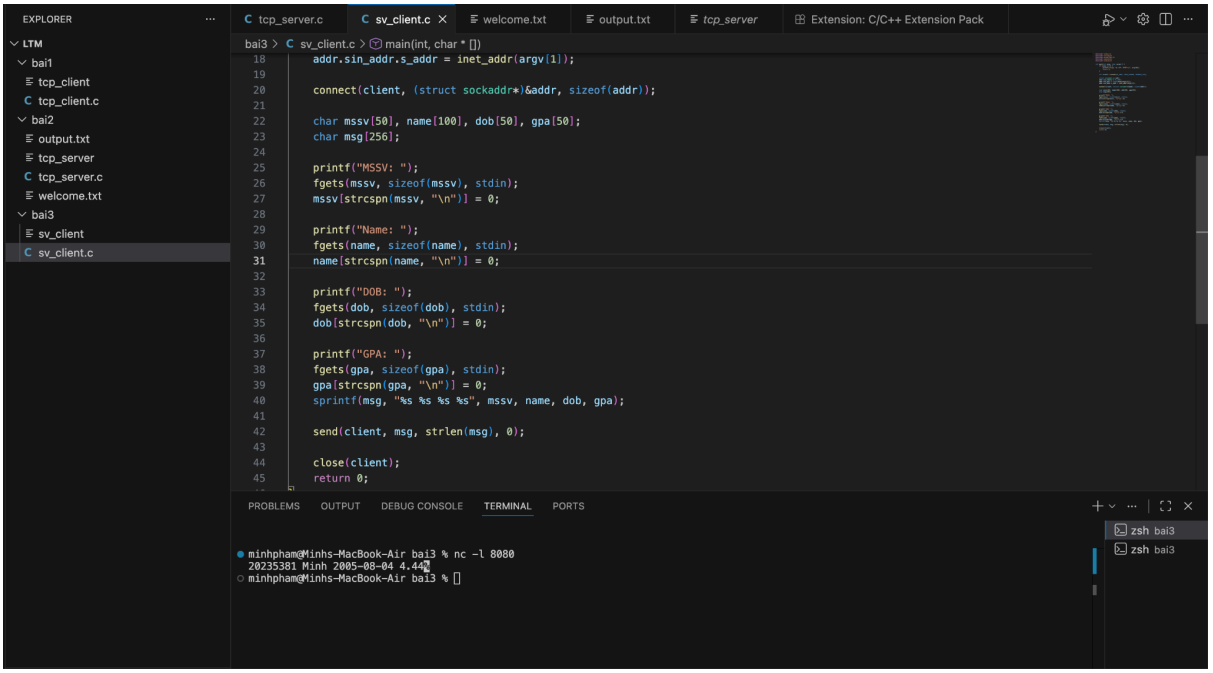
PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
minhpham@Minhs-MacBook-Air ltm % cd bai2
minhpham@Minhs-MacBook-Air bai2 % cat output.txt
hello
abc
123
minhpham@Minhs-MacBook-Air bai2 %
```

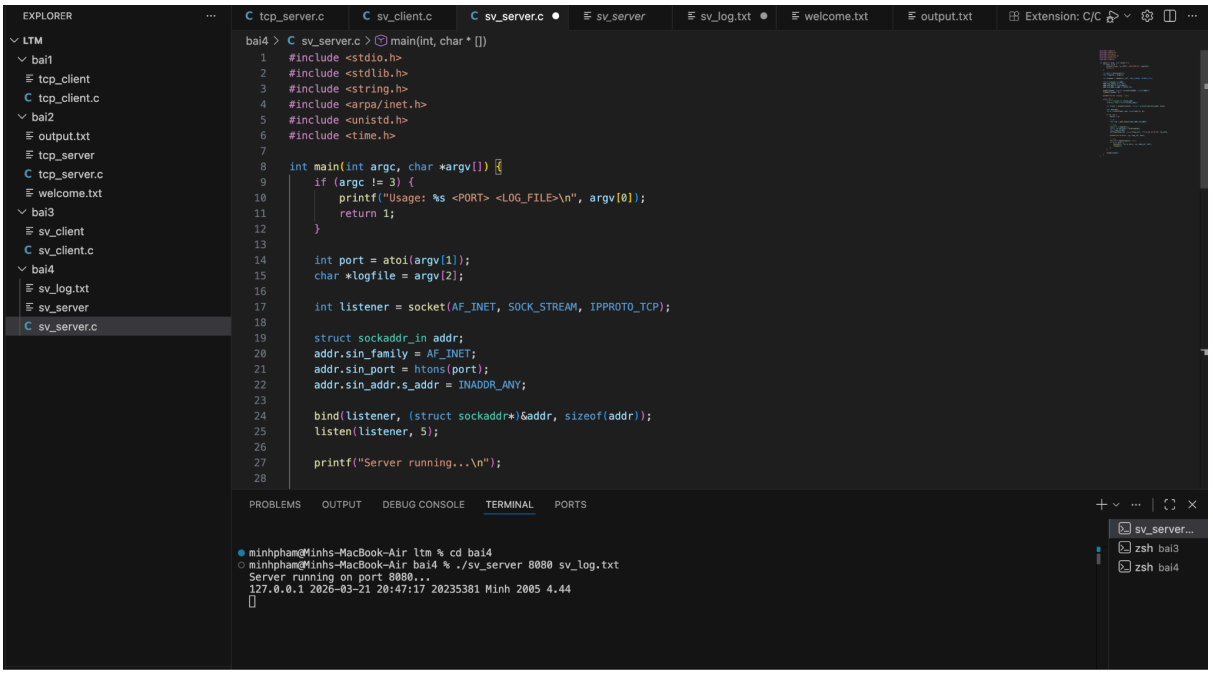
tcp_server... | tcp_client... | zsh...

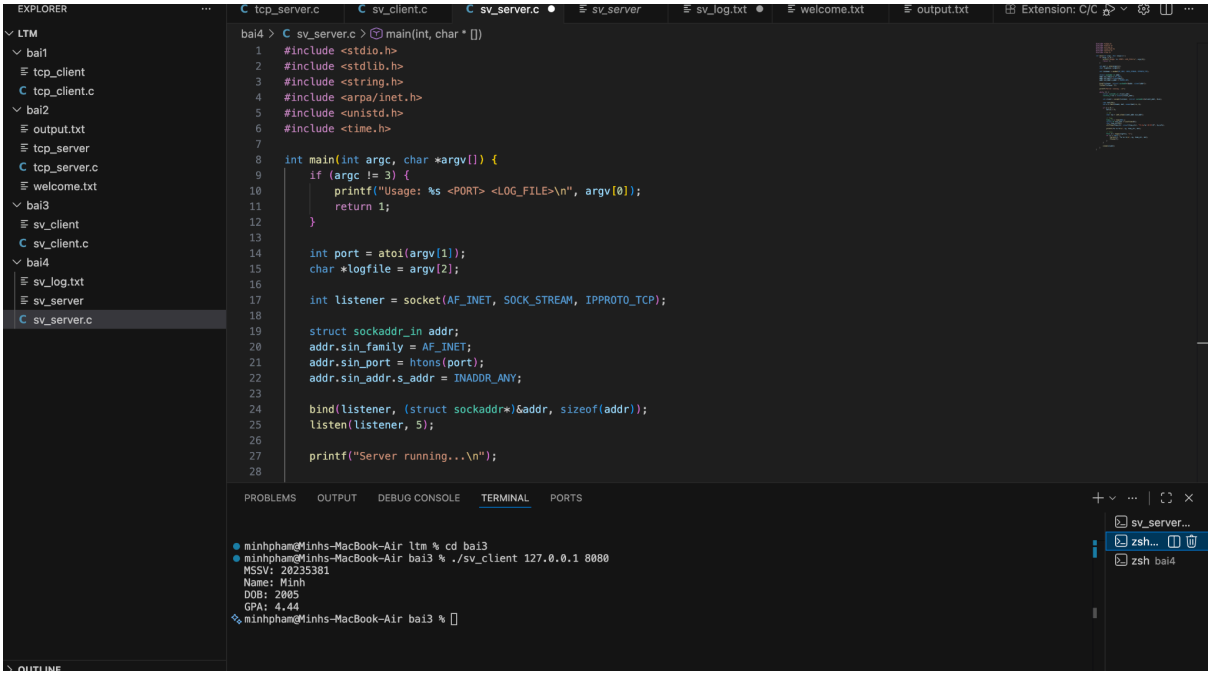
Bài 3:


EXPLORER
LTM
bai1
tcp_client
tcp_client.c
bai2
output.txt
tcp_server
tcp_server.c
welcome.txt
bai3
sv_client
sv_client.c
tcp_server.c
sv_client.c
welcome.txt
output.txt
tcp_server
Extension: C/C++ Extension Pack
bai3 > sv_client.c > main(int, char * [])
addr.sin_addr.s_addr = inet_addr(argv[1]);
connect(client, (struct sockaddr*)&addr, sizeof(addr));
char mssv[50], name[100], dob[50], gpa[50];
char msg[256];
printf("MSSV: ");
fgets(mssv, sizeof(mssv), stdin);
mssv[strcspn(mssv, "\n")] = 0;
printf("Name: ");
fgets(name, sizeof(name), stdin);
name[strcspn(name, "\n")] = 0;
printf("DOB: ");
fgets(dob, sizeof(dob), stdin);
dob[strcspn(dob, "\n")] = 0;
printf("GPA: ");
fgets(gpa, sizeof(gpa), stdin);
gpa[strcspn(gpa, "\n")] = 0;
sprintf(msg, "%s %s %s %s", mssv, name, dob, gpa);
send(client, msg, strlen(msg), 0);
close(client);
return 0;
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
minhpham@Minhs-MacBook-Air bai3 % ./sv_client 127.0.0.1 8080
MSSV: 20235381
Name: Minh
DOB: 2005-08-04
GPA: 4.44
minhpham@Minhs-MacBook-Air bai3 %
zsh bai3
zsh bai3
OUTLINE
TIMELINE
4 Hidden T...


EXPLORER
LTM
bai1
tcp_client
tcp_client.c
bai2
output.txt
tcp_server
tcp_server.c
welcome.txt
bai3
sv_client
sv_client.c
tcp_server.c
sv_client.c
welcome.txt
output.txt
tcp_server
Extension: C/C++ Extension Pack
bai3 > sv_client.c > main(int, char * [])
addr.sin_addr.s_addr = inet_addr(argv[1]);
connect(client, (struct sockaddr*)&addr, sizeof(addr));
char mssv[50], name[100], dob[50], gpa[50];
char msg[256];
printf("MSSV: ");
fgets(mssv, sizeof(mssv), stdin);
mssv[strcspn(mssv, "\n")] = 0;
printf("Name: ");
fgets(name, sizeof(name), stdin);
name[strcspn(name, "\n")] = 0;
printf("DOB: ");
fgets(dob, sizeof(dob), stdin);
dob[strcspn(dob, "\n")] = 0;
printf("GPA: ");
fgets(gpa, sizeof(gpa), stdin);
gpa[strcspn(gpa, "\n")] = 0;
sprintf(msg, "%s %s %s %s", mssv, name, dob, gpa);
send(client, msg, strlen(msg), 0);
close(client);
return 0;
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
minhpham@Minhs-MacBook-Air bai3 % nc -l 8080
20235381 Minh 2005-08-04 4.44
minhpham@Minhs-MacBook-Air bai3 %
zsh bai3
zsh bai3

Bài 4:


EXPLORER
LTM
bai1
tcp_client
tcp_client.c
bai2
output.txt
tcp_server
tcp_server.c
welcome.txt
bai3
sv_client
sv_client.c
bai4
sv_log.txt
sv_server
sv_server.c
tcp_server.c
sv_client.c
sv_server.c
sv_server
sv_log.txt
welcome.txt
output.txt
Extension: C/C
bai4 > sv_server.c > main(int, char * [])
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <arpa/inet.h>
#include <unistd.h>
#include <time.h>
int main(int argc, char *argv[]) {
if (argc != 3) {
printf("Usage: %s <PORT> <LOG_FILE>\n", argv[0]);
return 1;
}
int port = atoi(argv[1]);
char *logfile = argv[2];
int listener = socket(AF_INET, SOCK_STREAM, IPPROTO_TCP);
struct sockaddr_in addr;
addr.sin_family = AF_INET;
addr.sin_port = htons(port);
addr.sin_addr.s_addr = INADDR_ANY;
bind(listener, (struct sockaddr*)&addr, sizeof(addr));
listen(listener, 5);
printf("Server running...\n");
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
minhpham@Minhs-MacBook-Air ltm % cd bai4
minhpham@Minhs-MacBook-Air bai4 % ./sv_server 8080 sv_log.txt
Server running on port 8080...
127.0.0.1 2026-03-21 20:47:17 20235381 Minh 2005 4.44
sv_server...
zsh bai3
zsh bai4


EXPLORER
LTM
bai1
tcp_client
tcp_client.c
bai2
output.txt
tcp_server
tcp_server.c
welcome.txt
bai3
sv_client
sv_client.c
bai4
sv_log.txt
sv_server
sv_server.c
tcp_server.c
sv_client.c
sv_server.c
sv_server
sv_log.txt
welcome.txt
output.txt
Extension: C/C
bai4 > sv_server.c > main(int, char * [])
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <arpa/inet.h>
#include <unistd.h>
#include <time.h>
int main(int argc, char *argv[]) {
if (argc != 3) {
printf("Usage: %s <PORT> <LOG_FILE>\n", argv[0]);
return 1;
}
int port = atoi(argv[1]);
char *logfile = argv[2];
int listener = socket(AF_INET, SOCK_STREAM, IPPROTO_TCP);
struct sockaddr_in addr;
addr.sin_family = AF_INET;
addr.sin_port = htons(port);
addr.sin_addr.s_addr = INADDR_ANY;
bind(listener, (struct sockaddr*)&addr, sizeof(addr));
listen(listener, 5);
printf("Server running...\n");
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
minhpham@Minhs-MacBook-Air ltm % cd bai3
minhpham@Minhs-MacBook-Air bai3 % ./sv_client 127.0.0.1 8080
MSSV: 20235381
Name: Minh
DOB: 2005
GPA: 4.44
minhpham@Minhs-MacBook-Air bai3 %
sv_server...
zsh...
zsh bai4

EXPLORER

- LTM
  - bai1
    - tcp_client
    - tcp_client.c
  - bai2
    - output.txt
    - tcp_server
    - tcp_server.c
    - welcome.txt
  - bai3
    - sv_client
    - sv_client.c
  - bai4
    - sv_log.txt
    - sv_server
    - sv_server.c

tcp_server.c | sv_client.c | sv_server.c ● | sv_server | sv_log.txt ● | welcome.txt | output.txt | Extension: C/C

bai4 > sv_server.c > main(int, char * [])

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <arpa/inet.h>
#include <unistd.h>
#include <time.h>

int main(int argc, char *argv[]) {
    if (argc != 3) {
        printf("Usage: %s <PORT> <LOG_FILE>\n", argv[0]);
        return 1;
    }

    int port = atoi(argv[1]);
    char *logfile = argv[2];

    int listener = socket(AF_INET, SOCK_STREAM, IPPROTO_TCP);

    struct sockaddr_in addr;
    addr.sin_family = AF_INET;
    addr.sin_port = htons(port);
    addr.sin_addr.s_addr = INADDR_ANY;

    bind(listener, (struct sockaddr*)&addr, sizeof(addr));
    listen(listener, 5);

    printf("Server running...\n");

```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
minhpham@Minhs-MacBook-Air ltm % cd bai4
minhpham@Minhs-MacBook-Air bai4 % cat sv_log.txt
127.0.0.1 2026-03-21 20:01:52 20235381 Minh 2005 4.44
127.0.0.1 2026-03-21 20:46:03 20235381 minh 2005 4.44
127.0.0.1 2026-03-21 20:47:17 20235381 Minh 2005 4.44
minhpham@Minhs-MacBook-Air bai4 %
```

sv_server...
zsh bai3
zsh...

OUTLINE